Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/03/2026
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Temporal Nature of Cognitive and Visuospatial Brain Domain Changes During Long-Duration Low-Earth Orbit Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	06/01/2019	**End Date:**	09/30/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2033 per S. Mack-Phillips/JSC (Ed., 8/17/23) NOTE: End date changed to 12/31/2027 per NSSC information (Ed., 1/27/21)		
Key Personnel Changes/Previous PI:	Co-I Christopher W. Jones left the University of Pennsylvania in September 2025. Dr. Basner will assume his responsibilities until a replacement is found. The list of Co-Investigators now reflects the current team.		

COI Name (Institution):	Gur, Ruben Ph.D. (The Trustees of the University of Pennsylvania) Moore, Tyler Ph.D. (Trustees of Tufts College) Roalf, David Ph.D. (University of Pennsylvania) Stahn, Alexander Ph.D. (Charite - Universitatsmedizin Berlin, Germany (University of Pennsylvania)) Bell, Suzanne (NASA) Ivkovic, Vladimir (Mass General Brigham) Dev, Sheena (NASA BHP Lab)
Grant/Contract No.:	80NSSC19K1046
Performance Goal No.:	
Performance Goal Text:	
Task Description:	As part of the CIPHER project (Complement of Integrated Protocols for Human Exploration Research), we are investigating visuospatial brain domain changes and spatial cognition in the up to 30 CIPHER crewmembers assigned to 6-, and 12-month ISS missions as well as in an equal number of age-, sex- and education-matched ground controls. The experiment is part of an international project consisting of three experiments (lead by Drs. Basner, Stahn, and Ivkovic, respectively) with synergistic aims that are being carried out in a joint effort by NASA and DLR/ESA. Dr. Stahn's project specifically targets spatial cognition, how it relates to structural and functional brain changes and their molecular signatures. Given the impact of visuospatial brain domain changes on neurobehavioral functioning, operations and safety during LDM (i.e., docking, landing, navigating on the planetary surface), spatial cognition and its neural basis are a key concern for extended mission durations. Dr. Ivkovic's team will assess sleep, stress and immune responses, and operational performance leveraging the Canada arm simulator— Robotic on Board Trainer for Research (ROBoT-r). These projects are presented separately in other abstracts. This project (PI Dr. Basner) focuses on neurostructural and cognitive changes. Astronauts will each perform NASA's Cognition test battery before, during, and after their 6-, or 12-month ISS missions. Based on protracted performance decrements observed in Scott Kelly after his one-year mission, the last test bout will be performed 1 year after return to Earth. The data will be used to generate temporal profiles of human cognitive performance that will inform future long-duration deep space missions. They will also be combined with existing Cognition data collected by the PI and his team on the ISS and in several space analog environments on Earth as the basis for a normative database for long-duration space missions. All astronauts will undergo structural and functional Magnetic Resonance Imaging (MRI) before and immediately after the mission (R+5, R+30). They will perform a version of the Cognition test battery that was specifically designed for the MRI (i.e., fCog) in the scanner. We will correlate structural and functional changes observed during fMRI scans post-flight relative to pre-flight with in-flight changes in Cognition performance to establish the biological basis for any observed cognitive performance changes. We will also integrate the general cognitive performance data from Cognition with spatial navigation, learning, and memory data as well as with biomarkers collected in Project B (Spatial Cognition and Hippocampal Plasticity During Long-Duration Low-Earth Orbit Missions: HypoCampus in i1YMP) before, during, and after flight. The current plan is to investigate 15 astronauts on 6- or 12-month ISS missions as well as 15 age and sex matched controls, who will perform tests/scans at the same intervals as the matched astronaut but with a 2-month delay.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The three projects will deliver a highly unique and comprehensive set of integrated neuroimaging and neurocognitive tools for the evaluation and ultimately prevention of adverse effects on brain structure and function that lead to behavioral effects associated with exploration-type missions. As the Cognition test battery was developed for high-performing subject populations, this work will also translate to high performing populations on Earth (e.g., physicians, submariners).
Task Progress:	Cognitive testing: Two astronauts have completed data acquisition, four astronauts are in post-flight data acquisition, and one astronaut is in in-flight data acquisition. No control has completed data acquisition, four controls are in post-flight data acquisition, two controls are in pre-flight data acquisition, and one control group participant withdrew. So far, only two out of 60 Cognition tests are missing in control subjects (96.7% complete). Recruitment for a replacement control participant is ongoing. Neuroimaging: In astronauts, so far 145 out of 152 (95.4%) expected fMRI sequences were acquired. In controls, so far 78 out of 80 (97.5%) expected fMRI sequences were acquired. All structural scans were acquired in astronauts and controls. The overall quality of the scans is high.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Roalf D, Basner M, Beer JC, Shinohara RT, Ruparel K, Moore TM, Dinges DF, Stahn AC, Nasrini J, Hermosillo E, Ecker AJ, Prabhakaran K, Almuqbel MM, Smith MG, Jones CW, Johannes B, Gerlach DA, Healey B, van den Berg FP, Gunga H-C, Melzer TR, Taylor BV, Gur RC. "Transient gray matter decline during antarctic isolation: Roles of sleep, exercise, and cognition." npj Microgravity. 2025 Jul 11;11:39. https://doi.org/10.1038/s41526-025-00497-6 ; PubMed PMID: 40646003; PubMed Central PMCID: PMC12254294 , Jul-2025
Conference Materials (Downloadable)	Basner M, Stahn AC, Jones CW, Bell S, Ivkovic V, Ecker AJ, Ruparel K, Moore TM, Roalf D, Gur RC. "Temporal nature of cognitive and visuospatial brain domain changes during long-duration low-Earth orbit missions." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2026 2025_HRP_CIPHER_Project_A_Basner.pdf